Keshav Sharma

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EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Programming Languages: Python, Java, C++, C, SQL, JavaScript

Core Computer Science: DSA, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

Software & Application Development: FastAPI, Flask, React.js, Node.js, REST APIs, API Development, Middleware

Databases & Data Engineering: MySQL, SQLite, MongoDB, Redis, Data Warehousing, ETL Concepts, Data Processing

Data Science & Machine Learning: Pandas, NumPy, Scikit-learn, PyTorch, Feature Engineering, Statistical Analysis, Anomaly Detection, Model Evaluation

Cloud & Development Tools: Git, GitHub, Docker, Render, Vercel, VS Code, Jupyter Notebook

Testing & Engineering Practices: PyTest, Unit Testing, Debugging, Test-Driven Development, Performance Optimization

EXPERIENCE

VOIS for Tech University Engagement Program **|LINK** — ***Aug – Sep, 2025***

- Engineered a modular data processing pipeline to preprocess, analyze, and query two real-world datasets containing 8,500+ records using LLM-powered natural language interfaces.
- Designed and implemented automated validation and response extraction workflows, improving the consistency and reliability of generated outputs across diverse user queries.
- Optimized data processing by integrating preprocessing, prompt generation, response parsing, and result validation into a reusable workflow, reducing manual analysis effort and improving scalability.

PROJECTS

AI-Powered Expense Receipt Scanner & Analytics Platform **|LINK**

- Built a full-stack Python application using FastAPI, React, SQLAlchemy, and SQLite, integrating EasyOCR, OpenCV, and NLP to automate receipt processing and reduce processing time by 40%.
- Engineered and validated REST APIs with JWT authentication, Google OAuth, CRUD operations, analytics, budget tracking, and data export for reliable application workflows.
- Deployed the application on Render and Vercel, implementing end-to-end workflows across AI processing, backend services, database operations, and frontend validation.

RedisLite **|LINK**

- Engineered a Redis-inspired in-memory key-value database in Python with a custom RESP parser, multithreaded TCP server, thread-safe storage, O(1) LRU caching, and TTL expiration for concurrent request processing.
- Implemented Write-Ahead Logging (WAL), atomic log compaction, and automatic crash recovery to ensure durable persistence and data consistency under failures and concurrent workloads.
- Developed 46 PyTest tests covering concurrency, edge cases, and recovery scenarios; benchmarked 619K SET ops/sec in-memory and 2.8K ops/sec with WAL persistence to validate reliability and performance.

Financial Transaction Anomaly Detection & Audit Dashboard **| LINK**

- Engineered an end-to-end financial analytics pipeline processing 50K+ PaySim transactions, combining statistical outlier analysis, duplicate detection, vendor-velocity checks, HHI concentration analysis, and Isolation Forest for transaction-risk detection.
- Designed a star-schema SQLite data warehouse and composite 0–100 risk scoring framework, identifying 357 high/critical-risk transactions with end-to-end analysis completed in under 7 seconds.
- Built a 5-page Power BI audit dashboard with 20+ DAX measures to surface vendor risk, anomaly rates, transaction trends, and risk insights, supported by automated pipeline tests.

API Rate Limiter Library **|LINK**

- Built a production-ready Python rate-limiting library implementing five throttling algorithms, Redis integration, and reusable Flask/FastAPI middleware for reliable API traffic control.
- Designed a modular architecture with pluggable storage backends and thread-safe concurrent request handling, enabling flexible deployment across application environments.
- Developed 30+ PyTest tests covering concurrency, edge cases, and algorithm correctness, and packaged the library on PyPI for public installation and reuse.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional **|LINK** — **2026**